

Problem Set 2

POLS W4732

Fall 2026

Due October 2 before the TA section

Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1 (A twice-repeated game). *Consider the stage game*

		Player 2		
		<i>A</i>	<i>B</i>	<i>C</i>
Player 1	<i>A</i>	9, 9	1, 6	−2, 10
	<i>B</i>	6, 1	5, 5	1, 0
	<i>C</i>	10, −2	0, 1	2, 2

Total payoff is $u^1 + \delta u^2$, where $\delta \in (0, 1]$. The proposed symmetric strategy plays *A* in period 1; after (A, A) it plays *B*; after any other first-period outcome it mixes between *B* with probability q and *C* otherwise.

1. Write a separate incentive constraint for every distinct first-period unilateral deviation; do not assume which one binds.
2. Distinguish Nash equilibrium of the repeated game from subgame-perfect equilibrium. Evaluate the claim that the profile is automatically SPE because the continuation at $q = 1/6$ is a stage-game equilibrium.
3. Characterize the set of (δ, q) supporting Nash equilibrium and the subset supporting SPE.

Exercise 2 (A revenue-sharing agreement). At date 0, both players simultaneously accept (*A*) or reject (*R*) a revenue-sharing agreement. If either rejects, they play the following Prisoner's Dilemma in periods 1 and 2:

	<i>C</i>	<i>D</i>
<i>C</i>	2, 2	0, 3
<i>D</i>	3, 0	1, 1

If both accept, Player 1 receives share α and Player 2 receives $1 - \alpha$ of the sum of the two stage payoffs in each period. Player i pays enforcement cost k_i at date 0. Use (unnormalized) discounted payoff $u^1 + \delta u^2$ and subtract k_i once.

1. Describe the terminal histories without listing them and give their number when there are T post-agreement periods.
2. Derive both acceptance constraints and characterize the feasible set of (α, k_1, k_2) .
3. Explain when no division rule induces agreement.

Exercise 3 (Tit-for-Tat). In the infinitely repeated Prisoner's Dilemma, a proposed solution checks deviations only after cooperative histories and concludes that Tit-for-Tat is an SPE whenever $\delta \geq 1/2$.

1. Identify the missing histories and reconstruct the complete one-shot-deviation analysis.
2. Now change the strategy profile as follows: each player, after observing D in the previous period retaliates with probability $1/2$ and plays C otherwise. Under what conditions is this strategy profile is a SPE?

Exercise 4 (Minmax and credible punishment). Consider the infinitely repeated game

		Player 2		
		<i>A</i>	<i>B</i>	<i>C</i>
Player 1	<i>A</i>	4, 4	3, 0	1, 0
	<i>B</i>	0, 3	2, 2	1, 0
	<i>C</i>	0, 1	0, 1	0, 0

And consider the following strategy profile: Initial play is (B, B) . After every history, the public state is either Reward, Punishment 1, Punishment 2, Punishment 3, or Permanent Punishment. Reward starts from (B, B) and prescribes B to both players. If the previous play is not (B, B) , compliance requires transitioning to three periods of (C, C) ; complying to the three periods of punishment returns to Reward, while any deviation during punishment moves to Permanent Punishment, which prescribes (C, C) forever.

1. Represent this strategy using the graphic state-based representation employed in the slides, including every transition.
2. At every state, identify the most profitable one-shot deviation from the above strategy.

3. If it is not an SPE, prove that it is not by specifying a history and a profitable one-shot deviation, together with the associated payoff comparison.

Exercise 5 (Alternating offers with breakdown). *Players divide a unit surplus and are perfectly patient. After rejection of Player 1's offer, bargaining ends with payoffs (b_1, b_2) with probability p_1 ; after rejection of Player 2's offer, it ends with the same payoffs with probability p_2 .*

1. Recover the two stationarity equations and solve the unique stationary SPE division.
2. Verify that setting $p_1 = p_2 = p$ gives

$$v_1 = \frac{1 - b_2 + (1 - p)b_1}{2 - p}, \quad v_2 = \frac{1 - b_1 + (1 - p)b_2}{2 - p}.$$

3. Check the limits $p \rightarrow 0$ and $p \rightarrow 1$ and identify which payoff is more sensitive to each asymmetric breakdown probability.